UNIT 3
IoT Devices and Electrical Components

Complete, Easy-to-Understand Study Notes
Exam-Ready | With Formulas, Examples, Comparisons,
Practice Questions and Quick Revision

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1. Electronic and Electrical Circuits

Simple definition: An electronic circuit is a combination of electronic components and wires connected together to perform a useful task.

An electric circuit is a closed path through which electric current can flow. A circuit may contain only two components, such as a battery and a bulb, or it may contain many components.

Electronic circuits are used to:

- Produce a required current or voltage signal.
- Amplify a weak signal, meaning make it stronger.
- Transfer data from one place to another.
- Control devices such as motors, LEDs, sensors and communication systems.

Easy example: In a torch, the battery gives electrical energy, the switch controls the path, wires carry current, and the bulb converts electrical energy into light.

Flow of a circuit:

Input → Circuit → Output
Important exam point: A circuit must be complete for continuous current to flow. If the path is broken, current cannot flow.

2. SI Units and Measurement

Simple definition: SI units are standard units used to measure physical quantities such as time, length, mass and electric current.

SI stands for **Système International**. It is the modern form of the metric system and is used all over the world. Using common units helps everyone understand measurements in the same way.

Prefixes are added to SI units to show very large or very small values easily. For example, 1 kilo-ohm means 1,000 ohms, and 1 milliampere means 0.001 ampere.

Symbol	Unit Name	Quantity Measured
s	Second	Time
m	Metre	Length
kg	Kilogram	Mass
A	Ampere	Electric current
K	Kelvin	Thermodynamic temperature
mol	Mole	Amount of substance
cd	Candela	Luminous intensity

Important exam point: The SI unit of electric current is the ampere (A).

3. Electric Charge

Simple definition: Electric charge is the basic property of matter that causes it to experience an electric force.

All matter is made of atoms. An atom contains three main particles:

- **Electrons:** Negatively charged particles that move around the nucleus.
- **Protons:** Positively charged particles present in the nucleus.
- **Neutrons:** Particles with no electric charge, also present in the nucleus.

Electric charge is measured in **coulombs (C)**. One coulomb is a very large amount of charge.

The charge of one electron is negative and equal to:

$$e = -1.602 \times 10^{-19}\ C$$

A proton has the same amount of charge but it is positive. If an atom has equal numbers of protons and electrons, it is electrically neutral.

Rule of charges:

- Like charges repel each other.
- Opposite charges attract each other.

Easy example: Two positive charges push away from each other, just like the same poles of two magnets. A positive and a negative charge pull towards each other.

Common mistake: Do not confuse electric charge with electric current. Charge is the quantity of electricity, while current is the movement of charge.

4. Electric Current and Its Direction

Simple definition: Electric current is the flow of electric charge through a conductor.

In most electrical circuits, current is carried by electrons moving through a metal wire. A conductor is a material that allows electric charge to move easily, such as copper or aluminium.

For a sustained current to flow, two things are required:

- A complete or closed circuit.
- A voltage source, such as a battery, to push the charges.

Easy example: Think of water flowing through a pipe. The water is like electric charge, the pipe is like a wire, and the water pump is like a battery.

Direction of current:

There are two directions to remember:

- **Conventional current:** Flows from the positive terminal to the negative terminal of a battery.
- **Electron flow:** Electrons actually move from the negative terminal to the positive terminal.

Therefore, conventional current is opposite to the direction of electron movement.

Important exam point: Current is measured in amperes (A). One ampere means one coulomb of charge flowing per second.

5. Voltage and Power

5.1 Voltage

Simple definition: Voltage is the electrical pressure or energy required to move electric charge from one point to another.

Voltage is also called **potential difference**. It is measured in **volts (V)**.

The formula is:

$$V = W / Q$$

Where:

- V = voltage in volts (V)
- W = energy in joules (J)
- Q = charge in coulombs (C)

One volt means one joule of energy is required to move one coulomb of charge.

Easy example: A 9 V battery gives more electrical “push” than a 1.5 V battery. That is why a 9 V battery can make a bulb glow more brightly if the bulb is suitable.

5.2 Power

Simple definition: Power is the rate at which electrical energy is used or supplied.

Power is measured in **watts (W)**.

$$P = W / t$$

Where:

- P = power in watts
- W = energy in joules

- t = time in seconds

In electrical circuits, power can also be calculated as:

$$P = V \times I$$

Easy example: If a device uses 12 V and 2 A, then its power is $P = 12 \times 2 = 24$ W.

Common mistake: Voltage is not the same as power. Voltage is the push, while power tells how fast energy is being used.

6. Closed, Open and Short Circuits

These three terms describe whether current can flow safely in a circuit.

Circuit Type	Meaning	Current	Effect
Closed circuit	Complete path is available	Current flows	Device works normally
Open circuit	Path is broken or disconnected	Current = 0	Device does not work
Short circuit	Direct connection with almost no resistance	Very high current	Can damage circuit or cause fire

6.1 Closed Circuit

A closed circuit has a complete path for current. The battery, wires, load and return path are all connected properly.

6.2 Open Circuit

In an open circuit, there is a break in the path. Therefore, current cannot flow and the current is zero.

An open circuit has practically infinite resistance.

Example: When a switch is OFF, it creates an open circuit.

6.3 Short Circuit

A short circuit occurs when current finds an easy path with very low or zero resistance. This causes a very high current to flow.

A short circuit can overheat wires, damage components, or cause a fire. This is why fuses and circuit breakers are used.

7. Circuit Protection: Fuse and Circuit Breaker

Simple definition: A fuse and a circuit breaker are protective devices that stop excessive current from damaging a circuit.

They protect circuits from:

- Overcurrent, meaning more current than the circuit can safely handle.
- Overload, meaning too many devices drawing current at the same time.
- Short circuits.

A fuse contains a thin wire. When current becomes too high, the wire melts and breaks the circuit. A fuse must usually be replaced after it works.

A circuit breaker works like a switch. When current is too high, it trips and opens the circuit. It can usually be reset after the problem is removed.

Feature	Fuse	Circuit Breaker
Working	Wire melts due to excess current	Switch trips due to excess current
After operation	Usually must be replaced	Can usually be reset
Speed	Generally fast	Fast, depending on type
Use	Small appliances, electronics	Homes, industries, large systems

Important exam point: Both devices protect a circuit from damage caused by overcurrent or short circuit.

8. Alternating Current (AC) and Direct Current (DC)

Simple definition: DC flows in one direction only, while AC repeatedly changes its direction.

8.1 Direct Current (DC)

- Current flows in only one direction.
- Its magnitude remains constant in an ideal DC source.
- Batteries, solar panels and USB power supplies provide DC.
- Most small electronic devices internally use DC.

8.2 Alternating Current (AC)

- Current periodically reverses direction.
- Its magnitude changes continuously with time.
- Household electricity supply is AC.
- AC is suitable for long-distance power transmission.

Easy example: DC is like water flowing steadily in one direction through a pipe. AC is like water moving forward and backward repeatedly.

Important exam point: A battery provides DC, while wall sockets generally provide AC.

9. Electronic Components: Active and Passive

Simple definition: Electronic components are parts used to control electric current and produce a desired output in a circuit.

Main electronic components include resistors, capacitors, inductors, diodes and transistors.

9.1 Active Components

An active component can supply energy to a circuit or control the flow of energy.

- It can amplify signals.
- It generally has directionality.
- Examples: battery, transistor, integrated circuit.

9.2 Passive Components

A passive component uses, stores, or controls energy but does not generate energy.

- It cannot amplify a signal by itself.
- Examples: resistor, capacitor, inductor.
- Some passive components, such as diodes and polarized capacitors, must be connected in a particular direction.

Feature	Active Component	Passive Component
Energy	Can inject or control energy	Uses or stores energy
Amplification	Can amplify signals	Cannot amplify by itself
Examples	Battery, transistor, IC	Resistor, capacitor, inductor
Direction	May have directionality	Usually no directionality

Common mistake: A diode is passive, but it must be installed in a specific direction because it allows current mainly in one direction.

10. Cell and Battery

Simple definition: A cell or battery is a device that converts chemical energy into electrical energy.

A battery works like a chemical pump for electrons. Chemical reactions inside the battery create a voltage difference between its two terminals.

A battery has two terminals:

- **Anode:** Negative terminal in a typical discharging battery.
- **Cathode:** Positive terminal in a typical discharging battery.

The battery voltage depends on the chemicals used inside it. When a battery is connected to a circuit, it provides the energy needed to move charges through the circuit.

Easy example: A mobile phone battery stores chemical energy. When the phone is used, this chemical energy is converted into electrical energy to run the phone.

Important exam point: A battery is an active component because it supplies electrical energy to a circuit.

11. Ground and Multimeter

11.1 Ground or Earth

Simple definition: Ground is a reference point in a circuit considered to be at zero volts.

Voltage is always measured between two points. Therefore, we need a reference point. This reference point is called ground or earth.

For example, if a point in a circuit is at 5 V with respect to ground, it means its voltage is 5 V higher than the ground point.

11.2 Multimeter

Simple definition: A multimeter is an instrument used to measure voltage, current, resistance and sometimes other electrical quantities.

A multimeter has two probes:

- **Red probe:** Usually connected to the positive or higher-potential point.
- **Black probe:** Usually connected to the negative terminal or reference point.

The display shows the voltage of the red terminal with respect to the black terminal.

Example: If black probe is at 0 V and red probe is at 220 V, the multimeter shows +220 V. If black probe is at 220 V and red probe is at 20 V, it shows -200 V.

Important exam point: Voltage readings depend on the reference point chosen.

12. Resistor

Simple definition: A resistor is a passive two-terminal component that opposes the flow of electric current.

A resistor provides **electrical resistance**. Resistance is measured in **ohms**, written as the Greek symbol omega.

Resistors are used to:

- Limit current in a circuit.
- Protect components such as LEDs from excessive current.
- Divide voltage in a circuit.
- Control signal levels.

Easy example: A resistor is like a narrow part of a water pipe. It reduces the amount of water that can flow through the pipe.

Important exam point: A resistor is a passive component and its unit is the ohm.

13. Capacitor

Simple definition: A capacitor is a passive component that stores electrical energy in an electric field.

A capacitor has two terminals and provides **capacitance**. Capacitance is measured in **farads (F)**.

The main formula is:

$$Q = C \times V$$

Where:

- Q = charge stored, in coulombs
- C = capacitance, in farads
- V = voltage across the capacitor, in volts

A capacitor tries to resist sudden changes in voltage. It can supply current in either direction to keep voltage stable.

13.1 Capacitor with DC

- When an uncharged capacitor is connected to a DC source, it starts charging.
- At the beginning, it behaves almost like a short circuit because its voltage is zero.
- When fully charged, its voltage becomes equal to the source voltage.
- After full charging, it behaves like an open circuit for steady DC.

Easy example: A capacitor is like a small water tank. It can store water for some time and release it later.

Common mistake: A fully charged capacitor does not allow steady DC current to continue flowing through it; it behaves like an open circuit.

14. Inductor

Simple definition: An inductor is a passive component, usually a coil of wire, that stores energy in a magnetic field.

An inductor is also called a **coil, choke, or reactor**. It provides **inductance**, measured in **henry (H)**.

The voltage across an inductor is:

$$V = L \times (di / dt)$$

Where:

- V = voltage in volts
- L = inductance in henry
- di/dt = rate of change of current with time

An inductor opposes sudden changes in current. If current tries to change quickly, the inductor produces a voltage that opposes that change.

Functions of an inductor:

- Store energy in a magnetic field.
- Control or filter signals.
- Oppose sudden current changes.

14.1 Inductor with DC

When steady DC flows through an inductor, the current does not change. Therefore, the magnetic flux does not change, and the inductor behaves like a short circuit for ideal steady DC.

Easy example: An inductor is like a heavy flywheel. It does not like sudden changes in speed, just as an inductor does not like sudden changes in current.

Important comparison: A capacitor stores energy in an electric field, while an inductor stores energy in a magnetic field.

15. Electromagnetic Waves and Faraday's Law

15.1 Electromagnetic Waves

Simple definition: Electromagnetic waves are waves produced by vibrations between an electric field and a magnetic field.

Electromagnetic waves can travel through space without needing a wire. Examples include radio waves, microwaves, infrared waves, visible light, ultraviolet rays, X-rays and gamma rays.

In IoT, electromagnetic waves are important because wireless devices use radio waves to communicate.

15.2 Faraday's First Law

Simple definition: When a conductor is placed in a changing magnetic field, a voltage called EMF is induced in the conductor.

EMF means **electromotive force**. It is basically an induced voltage difference.

If the conductor is part of a closed circuit, the induced EMF causes current to flow.

Easy example: Move a magnet near a coil of wire. The changing magnetic field induces voltage in the coil. This is the basic principle behind generators and transformers.

Important exam point: A changing magnetic field is necessary to induce EMF. A steady magnetic field does not induce EMF.

16. Diode

Simple definition: A diode is a two-terminal component that allows current to flow mainly in one direction.

A diode works like a one-way street for current.

It has two ends:

- **Anode:** Positive terminal of the diode.
- **Cathode:** Negative terminal of the diode.

When the anode is more positive than the cathode, the diode is forward biased and current can flow. When the cathode is more positive than the anode, the diode is reverse biased and it blocks current.

16.1 Ideal Diode Behaviour

Condition	Ideal Diode Behaviour	Equivalent Circuit
Forward biased	Zero voltage drop, current flows	Short circuit / wire
Reverse biased	Current is zero	Open circuit

Easy example: A diode is like a one-way valve in a water pipe. Water can flow in one direction but not in the opposite direction.

Important exam point: A diode allows current mainly from anode to cathode.

17. Switch and Relay

17.1 Switch

Simple definition: A switch is a component that makes or breaks an electrical circuit.

- When the switch is ON, the circuit is closed and current can flow.
- When the switch is OFF, the circuit is open and current stops.
- Switches can be manual, such as a light switch, or automatic, such as a sensor-controlled switch.

17.2 Relay

Simple definition: A relay is an electrically operated switch.

A relay uses a small control signal to switch a larger electrical load. It has input terminals for the control signal and output contact terminals for the load circuit.

Easy example: A microcontroller gives a small signal to a relay. The relay then switches ON a high-power device such as a motor, pump, fan, or light.

Important exam point: A relay provides electrical isolation between the low-power control circuit and the high-power load circuit.

18. Ohm’s Law

Simple definition: Ohm’s law states that current through a conductor is directly proportional to voltage and inversely proportional to resistance.

The formula is:

$$V = I \times R$$

Other forms:

$$I = V / R$$

$$R = V / I$$

Where:

- V = voltage in volts
- I = current in amperes
- R = resistance in ohms

Solved example:

A circuit has a voltage of 12 V and resistance of 4 ohms. Find the current.

$$I = V / R = 12 / 4 = 3 \text{ A}$$

Answer: The current is 3 amperes.

Important exam point: If voltage increases and resistance remains the same, current increases. If resistance increases and voltage remains the same, current decreases.

19. Kirchhoff’s Laws

Kirchhoff’s laws are used to analyse electrical circuits.

19.1 Basic Terms

- **Node:** A point where two or more circuit components are joined.
- **Loop:** Any closed path in a circuit that does not pass through the same node more than once.
- **Mesh:** A loop that does not contain any other loop inside it.

19.2 Kirchhoff’s Current Law (KCL)

Simple definition: The total current entering a node is equal to the total current leaving that node.

$$\text{Algebraic sum of currents at a node} = 0$$

Current entering is taken as positive and current leaving is taken as negative.

Solved example: If 10 A enters a node and 3 A leaves through one branch, then the current in the other branch is:

$$I = 10 - 3 = 7 \text{ A}$$

19.3 Kirchhoff’s Voltage Law (KVL)

Simple definition: The total voltage around any closed loop in a circuit is zero.

$$\text{Algebraic sum of voltages around a loop} = 0$$

Voltage rises and voltage drops must balance each other.

Solved example: If a source voltage E = 12 V and one voltage drop V2 = 7 V, then the other voltage drop is:

V1 = 12 - 7 = 5 V

Easy example: KCL is like traffic at a road junction: cars entering must equal cars leaving. KVL is like walking around a hill: total upward movement equals total downward movement when you return to the start.

20. Series and Parallel Circuits

20.1 Series Circuit

Simple definition: In a series circuit, components are connected one after another, so there is only one path for current.

In a series circuit:

- The same current flows through all components.
- The total voltage is divided among the components.
- If one component becomes open, the entire circuit stops working.

For resistors in series:

R_total = R1 + R2 + R3 + ...

Example: If R1 = 2 ohms, R2 = 3 ohms and R3 = 5 ohms, then R_total = 2 + 3 + 5 = 10 ohms.

20.2 Parallel Circuit

Simple definition: In a parallel circuit, components are connected across separate branches, so there are multiple paths for current.

In a parallel circuit:

- The voltage across each branch is the same.
- The total current is divided among the branches.
- If one branch becomes open, the other branches can still work.

For current in parallel:

I_total = I1 + I2 + I3 + ...

For voltage in parallel:

V_total = V1 = V2 = V3

For resistors in parallel:

1 / R_total = 1 / R1 + 1 / R2 + 1 / R3

Example: If two resistors of 6 ohms each are connected in parallel:

1/R_total = 1/6 + 1/6 = 2/6 = 1/3

Therefore, R_total = 3 ohms.
Important exam point: In parallel, total resistance is always less than the smallest individual resistance.

21. Equivalent Resistance and Voltage Divider

21.1 Equivalent Resistance

Simple definition: Equivalent resistance is the single resistance that can replace a combination of resistors without changing the total current or voltage.

For series resistors, add the resistances directly. For parallel resistors, use the reciprocal formula.

21.2 Voltage Divider Rule

Simple definition: A voltage divider is a series circuit that divides a source voltage into smaller voltages.

For two series resistors R1 and R2 connected across voltage V:

$V1 = V \times R1 / (R1 + R2)$
$V2 = V \times R2 / (R1 + R2)$

Example: If V = 12 V, R1 = 4 ohms and R2 = 8 ohms:

$V1 = 12 \times 4 / 12 = 4$ V

$V2 = 12 \times 8 / 12 = 8$ V

Answer: The 4-ohm resistor has 4 V across it, and the 8-ohm resistor has 8 V across it.

22. Transformer

Simple definition: A transformer is a passive component that transfers electrical energy from one circuit to another using electromagnetic induction.

A transformer usually has two coils:

- **Primary coil:** Connected to the input AC voltage.
- **Secondary coil:** Provides the output AC voltage.

A transformer works only with AC because it needs a changing magnetic field to induce voltage in the secondary coil.

Types of transformers:

- **Step-up transformer:** Increases voltage from primary to secondary.
- **Step-down transformer:** Decreases voltage from primary to secondary.

Easy example: A mobile charger uses a transformer-type circuit to reduce high AC mains voltage to a lower, safer voltage before converting it to DC.

Important exam point: A transformer transfers energy but does not create energy.

23. Transistor and BJT

23.1 Transistor

Simple definition: A transistor is a small three-terminal semiconductor device used for switching and amplifying electrical signals.

The word transistor comes from “transfer” and “resistance”. It transfers resistance between the input and output parts of a circuit.

Uses of a transistor:

- Act as an electronic switch.
- Amplify weak signals.
- Control current in a circuit.
- Form part of integrated circuits and processors.

Types of transistors include BJT, MOSFET, phototransistor, Darlington transistor and power transistor.

23.2 BJT: Bipolar Junction Transistor

Simple definition: A BJT is a three-layer semiconductor device with two PN junctions, used for switching and amplification.

BJT usually contains silicon. Impurities are added to silicon through a process called doping to create p-type and n-type semiconductor regions.

A BJT has three terminals:

- **Emitter (E):** Sends charge carriers.
- **Base (B):** Controls the transistor.
- **Collector (C):** Collects charge carriers.

There are two main types of BJT:

- **NPN transistor:** Current flows from collector to emitter when the base is properly biased.
- **PNP transistor:** Current flows from emitter to collector when the base is properly biased.

The arrow in a transistor symbol shows the direction of conventional current flow.

Easy example: A small base current can control a much larger collector current. This is like a small tap controlling the flow of water in a large pipe.

Important exam point: A transistor is an active component because it can amplify signals.

24. Integrated Circuits, 555 Timer and Datasheet

24.1 Integrated Circuit (IC)

Simple definition: An integrated circuit is a small chip that contains many electronic components connected together on a semiconductor material, usually silicon.

An IC may contain hundreds to millions of transistors, resistors and capacitors.

Uses of ICs:

- Amplifiers.
- Oscillators.
- Timers.
- Counters.
- Logic gates.
- Computer memory.
- Microcontrollers and processors.

ICs are the basic building blocks of modern electronic devices.

24.2 555 Timer IC

Simple definition: The 555 timer IC is a popular integrated circuit used to generate time delays, pulses and oscillations.

It can be used in timer circuits, LED flashing circuits, tone generators and pulse-generation circuits.

24.3 Datasheet

Simple definition: A datasheet is a document that gives detailed technical information about an electronic component or IC.

A datasheet may include pin configuration, electrical ratings, operating voltage, current limits, timing diagrams and example circuits.

Important exam point: Before using an IC, its datasheet should be checked to avoid connecting it incorrectly or exceeding its ratings.

25. Microprocessor, Microcontroller and SoC

25.1 Microprocessor

Simple definition: A microprocessor is the central processing unit of a computer system, containing arithmetic, logic and control circuits on one or a few ICs.

A microprocessor performs calculations, makes decisions and controls the operation of a computer. However, memory and input/output devices are usually connected externally.

25.2 Microcontroller

Simple definition: A microcontroller is a small computer on a single chip that contains a CPU, memory and programmable input/output peripherals.

A microcontroller is used in embedded systems, where it controls a particular device automatically.

25.3 System on Chip (SoC)

Simple definition: A System on Chip is an integrated circuit that combines most components of a complete computer or electronic system on one chip.

An SoC may include a processor, memory controllers, graphics processing unit, communication circuits and other system components.

25.4 Comparison Table

Parameter	Microprocessor	Microcontroller
Definition	Heart of a computer system	Heart of an embedded system
Memory and I/O	Usually connected externally	Usually present inside the chip
Circuit complexity	More complex due to external components	Less complex and compact
Efficiency	Less efficient for small control tasks	More efficient for dedicated tasks
Registers	Generally fewer	Generally more
Applications	Personal computers, laptops	Washing machines, ACs, IoT devices

Easy example: A microprocessor is like the brain of a large computer. A microcontroller is like a small brain with memory and senses built into one tiny chip, used to control a washing machine.

Important exam point: A microcontroller is preferred for compact embedded and IoT systems because it includes CPU, memory and I/O on one chip.

26. Final Revision, Formulas and Practice Questions

26.1 Important Definitions

- **Electric charge:** Basic property of matter responsible for electric force; measured in coulombs.
- **Electric current:** Flow of electric charge; measured in amperes.
- **Voltage:** Potential difference or electrical pressure; measured in volts.
- **Power:** Rate of using or supplying energy; measured in watts.
- **Resistance:** Opposition to current flow; measured in ohms.
- **Capacitor:** Component that stores energy in an electric field.
- **Inductor:** Component that stores energy in a magnetic field.
- **Diode:** Component that allows current mainly in one direction.
- **Transistor:** Semiconductor device used for switching and amplification.
- **Relay:** Electrically operated switch.
- **Transformer:** Device that transfers AC electrical energy between circuits.
- **IC:** Integrated circuit containing many electronic components on one chip.
- **Microcontroller:** Small computer with CPU, memory and I/O on one chip.
- **SoC:** System on Chip that integrates most parts of an electronic system.

26.2 Important Formulas

- V = W / Q
- P = W / t
- P = V × I
- V = I × R
- I = V / R
- R = V / I
- Q = C × V
- V = L × (di / dt)
- Series resistance: R_total = R1 + R2 + R3 + ...
- Parallel resistance: 1/R_total = 1/R1 + 1/R2 + 1/R3
- Voltage divider: V1 = V × R1 / (R1 + R2)

26.3 Quick Comparison

Concept	Capacitor	Inductor
Stores energy in	Electric field	Magnetic field
Unit	Farad (F)	Henry (H)
Formula	Q = CV	V = L(di/dt)
Opposes change in	Voltage	Current
Behaviour with steady DC	Open circuit after charging	Short circuit

26.4 Last-Minute Checklist

- Know the SI units of charge, current, voltage, power and resistance.
- Remember that conventional current is opposite to electron flow.
- Understand open, closed and short circuits.

- Remember capacitor stores electric energy; inductor stores magnetic energy.
- Remember diode allows current mainly from anode to cathode.
- Apply Ohm's law correctly using V = IR.
- Use KCL at nodes and KVL around loops.
- Know the difference between series and parallel circuits.
- Know the difference between microprocessor, microcontroller and SoC.

26.5 Practice Questions

Short-answer questions:

- Define electric current and state its SI unit.
- What is voltage? Write its formula.
- State Ohm's law.
- What is the difference between AC and DC?
- What is a short circuit?
- What is the function of a diode?
- What is a relay?
- Differentiate between a microprocessor and a microcontroller.

Long-answer questions:

- Explain Ohm's law with formula, units and a solved example.
- Explain series and parallel circuits with formulas and examples.
- State and explain Kirchhoff's current law and voltage law.
- Explain the working of a capacitor and an inductor with DC.
- Explain the construction and working of a BJT.
- Compare microprocessor, microcontroller and SoC.

26.6 MCQs

1. The SI unit of electric current is:
A. Volt B. Ampere C. Ohm D. Watt
Answer: B. Ampere — current is measured in amperes.

2. The unit of resistance is:
A. Volt B. Farad C. Ohm D. Henry
Answer: C. Ohm.

3. Which component stores energy in an electric field?
A. Resistor B. Capacitor C. Inductor D. Diode
Answer: B. Capacitor.

4. Which component stores energy in a magnetic field?
A. Capacitor B. Resistor C. Inductor D. Battery
Answer: C. Inductor.

5. A diode allows current to flow mainly:
A. In both directions B. In one direction C. Only in AC D. Only when reverse biased
Answer: B. In one direction.

6. In a series circuit, which quantity remains the same?
A. Voltage B. Current C. Resistance D. Power
Answer: B. Current.

7. In a parallel circuit, which quantity remains the same across branches?
A. Current B. Voltage C. Resistance D. Power
Answer: B. Voltage.

8. Which device is an electrically operated switch?
A. Resistor B. Relay C. Capacitor D. Transformer
Answer: B. Relay.

9. Which device contains CPU, memory and I/O on one chip?
A. Microprocessor B. Microcontroller C. Transformer D. Diode
Answer: B. Microcontroller.

10. If V = 10 V and R = 5 ohms, then current is:
A. 2 A B. 5 A C. 50 A D. 0.5 A
Answer: A. 2 A, because I = V/R = 10/5.

End of Unit 3 Study Notes.